

Functions & Graphs

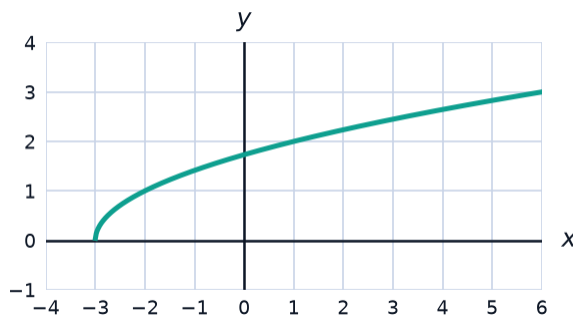


Evaluating functions

- 1 If $f(x) = 3x - 2$, find $f(4)$.
- 2 If $f(x) = x^2 + 2x$, find $f(-3)$.
- 3 Consider $f(x) = 2x^2 - 5x + 1$. Evaluate:
a $f(0)$ **b** $f(2)$

Domain and range

- 4 The graph of $y = \sqrt{x + 3}$ is shown below. State the domain and range based on the graph, then verify the domain algebraically.



- 5 Find the domain of $f(x) = \frac{1}{x - 5}$.
- 6 Find the domain of $f(x) = \sqrt{2x - 8}$.

Composition of functions

- 7 If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $(f \circ g)(2)$.
- 8 If $f(x) = x + 5$ and $g(x) = 3x$, find $(g \circ f)(x)$ as a simplified expression.
- 9 If $f(x) = x^2$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and state whether they are equal.

Inverse functions

- 10 Find the inverse of $f(x) = 4x - 8$.
- 11 Find the inverse of $f(x) = \frac{x-3}{2}$.
- 12 Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses of each other by computing $f(g(x))$.
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Transformations of graphs

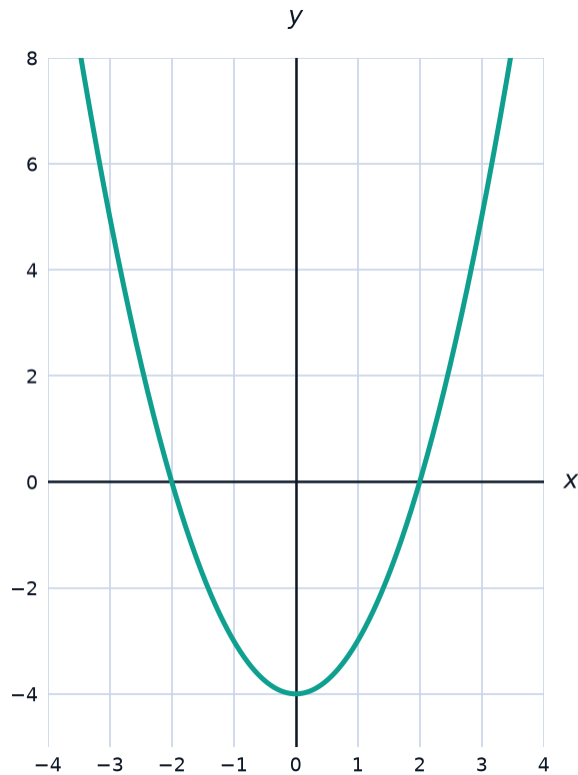
- 13 The graph of $y = f(x)$ is shifted 4 units left and 2 units down. Write the new function in terms of f .
- 14 If $f(x) = x^2$ and $g(x) = -f(x) + 3$, describe the transformation from f to g .
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Piecewise functions

- 15 A function is defined by $f(x) = x + 2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Find $f(-3)$ and $f(3)$.
- 16 A function is defined by $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(2)$ and $f(5)$, and state whether f is continuous at $x = 2$.
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Reading function graphs

- 17 The graph of $f(x) = x^2 - 4$ is shown below. Use the graph to estimate the x -intercepts and y -intercept, then find them exactly.



Even, odd, and other function properties

- 18 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, using $f(-x)$.
- 19 Determine whether $f(x) = x^3 - x$ is even, odd, or neither.

A well-designed word problem: modeling a car rental cost

- 20 This question has three parts. A rental car company models the cost of a rental as $C(d) = 45d + 20$ dollars, where d is the number of days rented. (a) Find $C(5)$, and interpret its meaning in context. (b) Define $P(d) = C(d) - 200$, representing profit relative to a fixed cost of 200 dollars per rental period. Write $P(d)$ explicitly and find $P(5)$. (c) Find the inverse function $C^{-1}(x)$, and use it to determine how many days someone rented the car if they were charged 335 dollars.

Multiple choice — easy

21 If $f(x) = -4x^2 + 3x - 4$, find $f(7)$.

a. -177

b. -179

c. -181

d. -11

22 If $f(x) = x^2 + 3x - 4$, find $f(-1)$.

a. -8

b. -3

c. -6

d. -4

23 If $f(x) = 3x^2 - 4x + 5$, find $f(6)$.

a. 87

b. 89

c. 91

d. -1

24 If $f(x) = -2x^2 + 5x - 2$, find $f(-3)$.

a. -11

b. -35

c. -33

d. -37

25 If $f(x) = x^2 - 9x$, find $f(8)$.

a. -64

b. -6

c. -8

d. -10

26 If $f(x) = -4x^2 - 5x - 6$, find $f(2)$.

a. -34

b. -30

c. -24

d. -32

27 If $f(x) = -4x^2 + 8x - 8$, find $f(4)$.

a. -42

b. -38

c. 8

d. -40

28 If $f(x) = 4x^2 + x - 9$, find $f(1)$.

a. -2

b. -4

c. -1

d. -6

29 If $f(x) = -3x^2 - 5x + 9$, find $f(-7)$.

a. -105

b. -103

c. 65

d. -101

30 If $f(x) = -4x^2 - 7x + 3$, find $f(8)$.

a. -309

b. -307

c. -311

d. -85

31 If $f(x) = -x^2 + x$, find $f(-1)$.

a. 0

b. -4

c. 1

d. -2

32 If $f(x) = -2x^2 - 3x + 1$, find $f(6)$.

a. -29

b. -87

c. -89

d. -91

33 If $f(x) = -3x^2 - 6x + 1$, find $f(-7)$.

- a. -102
- b. -106
- c. 64
- d. -104

34 If $f(x) = -4x^2 + 3x + 7$, find $f(8)$.

- a. -223
- b. -227
- c. -1
- d. -225

35 If $f(x) = -4x^2 - 5x + 6$, find $f(-1)$.

- a. 5
- b. 7
- c. 9
- d. 15

Multiple choice — medium

36 If $f(x) = x - 2$ and $g(x) = 4x^2 + 3$, find $(f \circ g)(-6)$.

- a. 148
- b. 147
- c. 145
- d. 143

37 If $f(x) = 3x + 1$ and $g(x) = x^2 + 4$, find $(f \circ g)(-1)$.

a. 16

b. 14

c. 18

d. 5

38 If $f(x) = -4x + 5$ and $g(x) = 3x^2 + 6$, find $(f \circ g)(-5)$.

a. -319

b. -317

c. 81

d. -321

39 If $f(x) = 4x - 3$ and $g(x) = 2x^2 + 2$, find $(f \circ g)(0)$.

a. 5

b. 3

c. 7

d. 2

40 If $f(x) = 2x - 8$ and $g(x) = x^2 + 5$, find $(f \circ g)(4)$.

a. 21

b. 36

c. 34

d. 32

41 If $f(x) = -3x - 9$ and $g(x) = -4x^2 + 2$, find $(f \circ g)(-3)$.

- a. 91
- b. -34
- c. 95
- d. 93

42 If $f(x) = -2x - 6$ and $g(x) = 2x^2 + 8$, find $(f \circ g)(-6)$.

- a. -164
- b. -166
- c. -168
- d. 80

43 If $f(x) = -4x + 3$ and $g(x) = -2x^2 - 5$, find $(f \circ g)(-1)$.

- a. 29
- b. 31
- c. 33
- d. -7

44 If $f(x) = -4x + 2$ and $g(x) = 2x^2 + 1$, find $(f \circ g)(0)$.

- a. 1
- b. -4
- c. 0
- d. -2

45 If $f(x) = 2x + 2$ and $g(x) = -3x^2 - 7$, find $(f \circ g)(-1)$.

- a. -20
- b. -18
- c. -10
- d. -16

46 If $f(x) = -x + 3$ and $g(x) = 2x^2 + 3$, find $(f \circ g)(-2)$.

- a. -8
- b. -10
- c. 11
- d. -6

47 If $f(x) = 2x + 2$ and $g(x) = x^2 + 4$, find $(f \circ g)(-3)$.

- a. 26
- b. 30
- c. 13
- d. 28

48 If $f(x) = 3x - 4$ and $g(x) = -2x^2 - 2$, find $(f \circ g)(1)$.

- a. -16
- b. -18
- c. -4
- d. -14

49 If $f(x) = -x + 9$ and $g(x) = 2x^2 + 7$, find $(f \circ g)(-4)$.

- a. -32
- b. -30
- c. 39
- d. -28

50 If $f(x) = -x - 3$ and $g(x) = 3x^2$, find $(f \circ g)(0)$.

- a. -1
- b. -5
- c. 0
- d. -3

Multiple choice — hard

51 Find the inverse of $f(x) = 3x - 7$.

- a. $\frac{x+7}{-3}$
- b. $\frac{x+7}{3}$
- c. $\frac{x-7}{3}$
- d. $3x - 7$

52 Find the inverse of $f(x) = 3x + 4$.

- a. $\frac{x-4}{-3}$
- b. $\frac{x-4}{3}$
- c. $\frac{x+4}{3}$
- d. $3x + 4$

53 Find the inverse of $f(x) = 4x - 2$.

a. $\frac{x+2}{-4}$

b. $\frac{x+2}{4}$

c. $4x - 2$

d. $\frac{x-2}{4}$

54 Find the inverse of $f(x) = -2x - 5$.

a. $-2x - 5$

b. $\frac{x-5}{-2}$

c. $\frac{x+5}{-2}$

d. $\frac{x+5}{2}$

55 Find the inverse of $f(x) = 4x + 9$.

a. $\frac{x-9}{-4}$

b. $\frac{x+9}{4}$

c. $\frac{x-9}{4}$

d. $4x + 9$

56 Determine whether $f(x) = 2x^3 + 9x$ is even, odd, or neither.

a. even

b. both even and odd

c. neither

d. odd

57 Determine whether $f(x) = 2x^4 + 7x^2$ is even, odd, or neither.

- a. both even and odd
- b. neither
- c. odd
- d. even

58 Determine whether $f(x) = x^2 + 4x$ is even, odd, or neither.

- a. even
- b. both even and odd
- c. neither
- d. odd

59 Determine whether $f(x) = -x^3 + 2$ is even, odd, or neither.

- a. both even and odd
- b. even
- c. neither
- d. odd

60 Determine whether $f(x) = -2x^2 + 8$ is even, odd, or neither.

- a. both even and odd
- b. odd
- c. even
- d. neither

- 61 A function is defined by $f(x) = 5x + 9$ for $x < 5$ and $f(x) = -3x + 5$ for $x \geq 5$. Find $f(2)$ and $f(5)$.
- a. 19, -10
 - b. 21, -10
 - c. -10, 19
 - d. 19, -12
- 62 A function is defined by $f(x) = -x$ for $x < -3$ and $f(x) = -5x - 5$ for $x \geq -3$. Find $f(-4)$ and $f(2)$.
- a. 6, -15
 - b. 4, -17
 - c. 4, -15
 - d. -15, 4
- 63 A function is defined by $f(x) = x + 3$ for $x < 0$ and $f(x) = x + 1$ for $x \geq 0$. Find $f(-1)$ and $f(0)$.
- a. 1, 2
 - b. 2, -1
 - c. 2, 1
 - d. 4, 1
- 64 A function is defined by $f(x) = 2x + 1$ for $x < 0$ and $f(x) = -3x - 6$ for $x \geq 0$. Find $f(-5)$ and $f(3)$.
- a. -9, -15
 - b. -9, -17
 - c. -7, -15
 - d. -15, -9

65 A function is defined by $f(x) = -3x + 7$ for $x < -5$ and $f(x) = 5x - 9$ for $x \geq -5$. Find $f(-9)$ and $f(-2)$.

a. 36, -19

b. 34, -19

c. -19, 34

d. 34, -21